

MID

Course: BTECH (123)

Set: B | Time: 2 Hours | Max Marks: 56 | Date: 15-03-2026

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

GENERAL INSTRUCTIONS:
HII

SECTION A: MCQs

1. Which quality component of usability refers to how easy it is for users to accomplish basic tasks the first time they encounter a design? **[3 Marks]**

- a) Efficiency
- b) Memorability
- c) Learnability
- d) Satisfaction

2. At what distance from the point of eye fixation is relative visual acuity approximately halved? **[3 Marks]**

- a) 1.0 degree
- b) 2.5 degrees
- c) 5.0 degrees
- d) 10.5 degrees

3. According to the text, what is the average reading speed for English prose on a computer monitor? **[3 Marks]**

- a) 300 words per minute
- b) 250 words per minute
- c) 200 words per minute
- d) 180 words per minute

4. Which psychological response to poor design occurs when road blocks prevent a task from being completed promptly and efficiently? **[3 Marks]**

- a) Confusion
- b) Annoyance
- c) Boredom
- d) Panic

5. Which of the following is a physical response to poor design where the user relies on an intermediary person to interact with the computer? **[3 Marks]**
- a) Modification of the task
 - b) Partial use of the system
 - c) Indirect use of the system
 - d) Compensatory activity
6. Based on the text, which group of users is more likely to scroll a page at a time rather than read line by line? **[3 Marks]**
- a) Young adults (18-36)
 - b) Expert programmers
 - c) Older adults (64-81)
 - d) Mandatory users

SECTION B: SHORT ANSWERS

7. According to the text, how is usability defined in terms of human performance? **[2 Marks]**
8. What are the two components that make up memory as described in the source text? **[2 Marks]**
9. Briefly explain Fitts' Law and its primary implication for screen design. **[2 Marks]**
10. Define foveal and peripheral vision as they relate to human interaction with a screen. **[2 Marks]**
11. What are the two levels of information processing that function simultaneously within a human? **[2 Marks]**

SECTION C: LONG ANSWERS

12. List and explain four common pitfalls in the development process that result from a flawed design process. **[4 Marks]**
13. Describe the characteristics of a "discretionary user" as compared to a "mandatory user." **[4 Marks]**
14. Explain four perceptual characteristics (Proximity, Similarity, Matching Patterns, and Closure) and their influence on design. **[4 Marks]**
15. Compare the traits of "Expert" users versus "Novice" users regarding their conceptual models and information processing. **[4 Marks]**

- 16.** Identify at least six common usability problems in graphical systems as reported by IBM specialists according to Mandel. **[4 Marks]**
- 17.** What are the five quality components of usability suggested by Nielsen (2003)? Provide a brief description **[4 Marks]**
- 18.** Gould (1988) suggests several techniques for gaining an understanding of users. Explain four of these methods **[4 Marks]**

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